

ASSIGNMENT - 2

Q-1 Describe the suitable method that can be used to measure a high VSWR.

Ans When VSWR is very high, direct measurement becomes inaccurate because the minima are extremely sharp and hard to detect. So, a more reliable technique that is Double Minima Method is ~~is~~ used.

Instead of measuring maximum & minimum voltages directly, this method measures the distance two points around a minimum where point is twice the minimum value.

Procedure -

- (1) Use a slotted line setup with movable probe.
- (2) Locate a minimum voltage point on the standing wave pattern.
- (3) Move the probe on both sides of this minimum.
- (4) Find the two points where the detected power is twice the minimum power (+3dB).
- (5) Measure the distance between these two points $\rightarrow$ call it d .

$$VSWR = \frac{1}{\frac{\lambda}{4d}}$$

$$VSWR = \frac{\lambda}{\pi d}$$

Q-2 Describe the different methods used for measurement of antenna gain.

- (1) Standard Antenna Method - This method uses two sets of measurements with the test and standard gain antennas.

Using test antenna of Gain G_T , Power Received is P_R

using standard antenna of Gain G_S , Power Received is P_S .

Then, $P_R/P_S = G_T/G_S$

$$G_T(\text{dB}) = G_S(\text{dB}) + 10 \log_{10} \left(\frac{P_R}{P_S} \right)$$

(2) Two Antenna Method- In this method, the signal transmitted

from a transmitting antenna of gain ~~G_T~~ G_t , and the signal is received by the test antenna of gain G_T , placed at a far-field distance R . The received power is expressed by-

$$P_R = \frac{P_t G_t G_T \lambda^2}{(4\pi R)^2}$$

$$G_T(\text{dB}) + G_t(\text{dB}) = 20 \log \left(\frac{4\pi R}{\lambda} \right) + 10 \log \left(\frac{P_R}{P_t} \right)$$

(3) Three Antenna Method- The three antenna method is the most general method to find gain of all the three antennas.

$$G_1(\text{dB}) + G_2(\text{dB}) = 20 \log \left(\frac{4\pi R}{\lambda} \right) + 10 \log \left(\frac{P_{R1}}{P_{t1}} \right)$$

$$G_2(\text{dB}) + G_3(\text{dB}) = 20 \log \left(\frac{4\pi R}{\lambda} \right) + 10 \log \left(\frac{P_{R2}}{P_{t2}} \right)$$

$$G_3(\text{dB}) + G_1(\text{dB}) = 20 \log \left(\frac{4\pi R}{\lambda} \right) + 10 \log \left(\frac{P_{R1}}{P_{t3}} \right)$$

Q.3 Describe the different methods used for the measurement of the frequency of a signal.

(1) Wave meter Method- A cavity resonator absorbs maximum power at its resonant frequency. When resonance

occurs, power dip is observed in the meter. We have cylindrical cavity with movable plunger, if plunger is adjusted, cavity length changes which changes resonant frequency.

(2) Slotted Line Method - In slotted line, probe is used to find two successive minima. The distance between two successive minima is d_{min} . $\lambda_g = 2(d_{min})$

Using formula, $\frac{1}{\lambda_0^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$ we calculated λ_0 ,

and hence, $f = \frac{c}{\lambda_0}$ which is required frequency.

(3) Down-converter Method - It uses Heterodyne principle, i.e. unknown frequency is mixed with a known frequency to produce a lower intermediate frequency (IF).

Hence, Final frequency,

$$f_x = f_a \pm f_{if}$$

↓
oscillator frequency (known)

Q.4. Calculate the SWR of a transmission system operating at 10 GHz. Assuming TE₁₀ wave transmission inside a waveguide of dimension $a=4\text{cm}$, $b=2.5\text{cm}$. The distance between two minimum power points = 1mm on a slotted line.

$$\Rightarrow f_0 = 10\text{GHz},$$

$$\lambda_0 = \frac{c}{f} = \frac{3 \times 10^8 \times 100}{10 \times 10^9} = 3\text{cm}$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{1 - \left(\frac{\lambda_0}{2a}\right)^2}} = \frac{3}{\sqrt{1 - \left(\frac{3}{2 \times 4}\right)^2}} = \frac{3}{\sqrt{1 - (0.375)^2}} = \frac{3}{\sqrt{1 - 0.1406}} \approx \frac{3}{0.927}$$

$$\boxed{\lambda_g \approx 3.24\text{cm}}$$

$$V_{SWR} = \frac{\lambda_g}{\pi d} = \frac{3.24}{\pi \times 0.1}$$

$$\boxed{V_{SWR} \approx 10.3} \quad \underline{\underline{\text{Ans}}}$$

Q-5. How are the microwave measurements different from low frequency measurements?

- (1) Circuit Behavior- At low frequency, lumped parameters are present on the other hand at microwave frequencies distributed parameters are present. That's why wavelength becomes comparable to circuit dimensions.
- (2) Measurement of Voltage & current- At microwave frequencies, direct measurement of voltages & current is not possible. Instead we measure VSWR, power, Reflection coefficient.
- (3) Power Loss & Radiation- At low frequency negligible radiation loss but at microwave frequencies significant radiation & dielectric losses are seen.

Q-6. Describe the technique of measuring the phase shift provided by network.

Ans we cannot measure phase directly at microwave frequencies, so we convert phase difference into a measurable distance using the slotted line technique.

Phase shift causes a shift in the standing wave pattern. This shift is observed as a displacement of minima/maxima position.

Without DUT find the position of minima = d_1 .

With ~~out~~ DUT find the position of minima = d_2 .

Phase shift,

$$\phi = \frac{2\pi}{\lambda_g} \Delta d$$

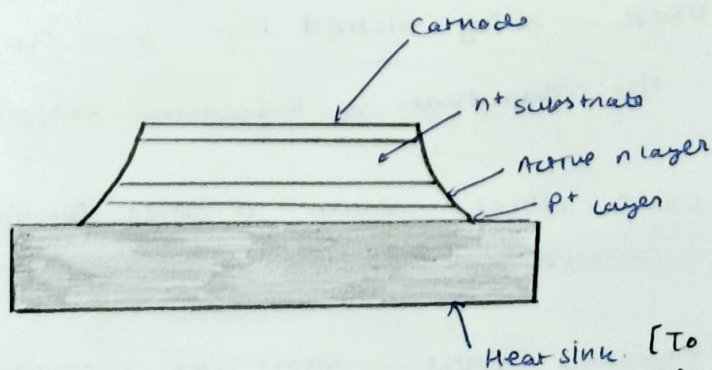
where, $\Delta d = d_2 - d_1$

Q-2. Describe the structure of the IMPATT diode with suitable diagram. Explain how negative resistance is achieved in an IMPATT diode.

Ans IMPATT (Impact Ionization Avalanche Transit Time) diodes rely on the effect of carrier impact ionization breakdown across a reverse biased P-N junction and the drift of carrier in the field (high field) region of a semiconductor to produce differential negative resistance.

The semiconductor layer structure contains $p^+ - n - n^+$ mounted on heat sink.

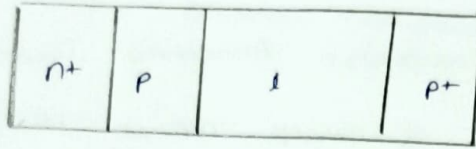
- (1) p^+ layer - This layer is heavily doped. Located at the bottom near heat sink and acts as anode. Form P-N junction with the n-layer. This junction is where the avalanche breakdown occurs.
- (2) Active n-layer - It is lightly doped. Also called drift region. The carriers generated in avalanche region drift through this layer. Thickness is carefully designed to control transit time.
- (3) n^+ substrate - It is also heavily doped. Provide low resistance path & act as a cathode. Ensure efficient current flow.



[To remove large heat generated due to high electric field & avalanche process]

Another structure of IMPATT diode i.e. Read diode has

$n^+ - p - i - p^+$



This additional i layer acts as drift region & provides transmit time delay & controls operating frequency.

To Achieve Negative Resistance-

- (1) The diode is reverse biased just below the breakdown voltage V_B . Also a small AC signal superimposed on DC bias.
- (2) During the half cycle (0 to $T/2$) total voltage exceeds V_B this causes avalanche multiplication. Here, current is delayed by $T/4$ (90°).
- (3) A short current pulse is generated at the junction. This pulse is already delayed by 90° .
- (4) Carriers move through i -region (drift region). $t_d = \frac{l}{v_d}$ and $t_d = T/2$

This adds another 90° delay.

This total phase shift $90^\circ + 90^\circ = 180^\circ$ leads to Negative Resistance in IMPATT diode.

Q.8. Explain the measurement of impedance using Smith chart.

Step 1- Measure VSWR - using slotted line find the VSWR S.

This gives the magnitude of reflection coefficient $|r|$.

Step 2- Draw VSWR circle - Draw a circle centered at origin. Radius corresponds to VSWR.

Step 3- Locate Reference Point - Short ckt corresponds to leftmost point of Smith chart.

Step 4- Move Toward Load - Move from short cut point toward load. Distance travelled = $\frac{d_{min}}{\lambda_g}$

Step 5- Locate the load point. That is normalized impedance Z_L .

Step 6- Then $Z_L = Z_0 \tilde{Z}_L$ Final impedance of load.

Q.9. Explain attenuation measurement in the microwave system.

Ans When a device or network is inserted in the transmission line, part P_r of the input power P_i is reflected from the input terminal & remaining part $P_i - P_r$ which actually enters the network is attenuated due to non-zero loss of the network.

$$\text{Attenuation (dB)} = 10 \log \left(\frac{P_{in}}{P_{out}} \right)$$

Step 1- Set microwave source frequency. Apply 1 kHz square wave modulation. Adjust for max. readings on VSWR meter.

Step 2- Ensure detector operates in square law region. Adjust input power to get ~ 30 dB reading. Set Reference $P_i = 1$ (0 dB)

Step 3- Measure frequency by using cavity wave meter.

Step 4- Measure the return loss,

$$\text{Return loss} = 10 \log (P_i / P_r)$$

Step 5- Measure the Insertion loss,

$$\text{Insertion loss} = 10 \log (P_i / P_o)$$

Step 6- Calculate Attenuation,

$$\text{Attenuation} = \text{Insertion loss} - \text{Reflection loss}$$

Q.10. What is Reflection coefficient? How it is measured?

Ans. The Reflection coefficient (denoted by Γ) is the ratio of the reflected wave amplitude to the incident wave amplitude at a load.

$$\Gamma = \frac{V_{\text{reflected}}}{V_{\text{incident}}}$$

In Terms of Impedance,

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Z_L = Load impedance

Z_0 = characteristic impedance

Slotted Line Method-

Step 1- Measure VSWR S using the slotted line.

Step 2- Calculating the magnitude of reflection coefficient $|\Gamma|$.

$$|\Gamma| = \frac{S-1}{S+1}$$

Step 3- Measure the position of Minima to find $d_{\min}$.

Step 4- Calculate phase angle,

$$\Phi = 2\beta d_{\min} - \pi$$

$$\beta = \frac{2\pi}{\lambda_g}$$

Step 5- Final Reflection coefficient,

$$\Gamma = |\Gamma| e^{j\Phi}$$